

Power Markets Casebook

v0.46.2

Applied decision cases for turning market, grid, contract, and portfolio concepts into professional decision memos and models.

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The Casebook gives bounded commercial exercises. Each case asks for a concrete artifact: a quote stack, duration test, site assessment, partner scorecard, product memo, or operating-model choice.

[**Download Casebook PDF**](#)[**Open Course**](#)[**Open Commercial Workbench**](#)

Methodology

These notes were built with a learning method designed for entering technical domains quickly: gather strong primary sources, identify the core guiding questions, work back to first principles, and build intuition through orders of magnitude, scaling relationships, and worked decision frames.

The workflow was accelerated by a custom AI research and knowledge-management system I built for semantic retrieval, source organization, drafting, and the editorial loops.

These notes are evolving. As I learn more and receive feedback, I keep improving the material.

Feedback is very welcome. Select any passage and click 'Send feedback' to send me your input on the spot.

How To Use The Casebook

Pick a case when the corresponding lecture idea feels familiar enough to use. If the answer feels hand-wavy, return to the linked lectures and rebuild the artifact.

- Read the decision question before the scenario data.
- Build the requested artifact before reading the worked solution.
- Compare your recommendation to the assumption flip.

Casebook Structure

The cases move from commercial memos to operating-model choices. Each case is standalone, and together they test whether the concepts can produce usable work.

Casebook

Applied Decision Practice

The cases give bounded practice in memos, scorecards, product tests, and operating-model choices.

- Use the given facts before adding judgment.
- Find the binding constraint and the first assumption that changes the answer.
- Produce a reusable artifact that records the decision logic.

[**Start Case 1**](#)[**Q&A Overview PDF**](#)

Glossary

Glossary

An alphabetical reference for the terms, acronyms, market roles, and technical objects used throughout the notes.

- Use it for quick recall.
- Return to the relevant lecture when the term needs mechanism, economics, or context.

[Open Glossary](#)

References

References

Books, reports, official pages, and selected practitioner analyses used across the notes.

- Market figures and rule-specific details include dates where timing matters.

[Open References](#)

Sofiane Boucenna, 2026

[in](#)

[Email](#)

Applied Cases

Case 1. Shaped Solar-Storage Power Purchase Agreement Quote

References: [1] [2] [3] [4] [5] [6]

A buyer asking for “solar plus storage” is usually asking for several things at once: price certainty, evidence of clean supply, a volume shape, and someone to carry the mismatch between production and consumption. The seller can accept that mismatch, sell a simpler product, or price the shape explicitly. The work product is a quote stack and a decision memo that make the residual obligation visible before the term sheet turns it into risk.

Decision Question

Should a renewable independent power producer (IPP) offer a shaped solar-plus-battery energy storage system (BESS) power purchase agreement (PPA), a simpler as-produced PPA, or no-bid?

Situation

A corporate buyer wants a 10-year clean-power product for a stable industrial load. The buyer prefers a flat hourly shape because its procurement team wants budget certainty and less exposure to evening prices.

The seller owns a solar project and can add a co-located battery. The commercial team asks for a quote that is attractive to the buyer, acceptable to investment committee, and explicit about who owns residual risk.

Scenario Data

These are scenario numbers for the case, not market quotes.

Item	Value
Buyer annual demand	100 GWh/year
Product requested	100 gigawatt-hours (GWh)/year flat profile, about 11.4 megawatts (MW) average
Solar project output	90 GWh/year

Item	Value
Co-located <u>BESS</u>	25 MW / 50 megawatt-hours (MWh)
Battery <u>round-trip efficiency</u>	88%
As-produced solar offer floor	45 EUR/MWh
Buyer maximum acceptable shaped price	68 EUR/MWh
Stress firming premium in hard hours	120 EUR/MWh
Residual hard-hour volume after solar+ <u>BESS</u>	20 MWh/h
Number of hard hours in stress case	250 h/year

For the base quote, use these illustrative quote-stack inputs. They are scenario assumptions for the exercise, not market benchmarks.

Quote component	EUR/MWh
Raw as-produced clean energy	45.0
Replacement clean energy and certificates for annual shortfall	1.0
Shape and <u>firming cost</u>	9.0
Expected <u>imbalance</u> and forecast error	2.0
Grid, tariff, and settlement administration	1.5
Certificates and evidence handling	1.0
Credit and tenor allowance	1.5
Battery <u>opportunity cost</u>	3.0

Quote component	EUR/MWh
Risk margin	2.0

For the dispatch appendix, use this synthetic annual summary. It is a compressed exercise input, not an 8,760-row market [backtest](#).

Dispatch item	Value
Average market price	58 EUR/MWh
PV-only capture price	45 EUR/MWh
PV+BESS capture price after dispatch	54 EUR/MWh
BESS energy charged from solar	12,000 MWh/year
BESS energy discharged after losses	10,560 MWh/year
Contract-sized shape test	100 GWh/year flat profile

Commercial Workbench Companion

Use the [Commercial Workbench](#) alongside this case to explore the same logic interactively: how much delivery is covered, where gaps remain, what indicative premium is being tested, when the battery may be worth more elsewhere, and what evidence you would want before taking the idea into a real project.

Terms To Pin Down Before Pricing

Before naming a number, the seller needs to pin down what the offer actually promises. A shaped [PPA](#) bundles clean MWh with settlement, evidence, and fallback obligations.

Term	Case Boundary	Why It Matters
Quantity	about 11.4 MW flat hourly profile, 100 GWh/year commercial target	fixes the shape being sold
Physical source	seller solar output plus co-located <u>BESS</u> where available	separates asset-backed delivery from market firming
Settlement	fixed shaped price against the agreed hourly profile	determines who pays when production and load differ
Residual firming	seller covers expected residuals; hard-hour tail must be capped, indexed, or passed through	prevents an uncapped short-volatility position
<u>Imbalance</u>	balance responsible party (<u>BRP</u>)-managed, with agreed data and performance reporting	connects contract economics to operational evidence
Certificates and matching evidence	defined certificate vintage, geography, delivery period, and data package	avoids selling a clean-energy claim without proof rules
Dispatch rights	seller/optimizer retains rights needed to deliver the shape and manage the <u>BESS</u>	stops the same flexibility being promised elsewhere
Data and audit rights	hourly meter data, certificate evidence, residual procurement records, dispatch logs, and settlement files are deliverable	makes the product testable after delivery
Credit and <u>collateral</u>	buyer <u>credit support</u> and seller exposure limit stated in the term sheet	prevents tenor and <u>collateral</u> from being buried in the energy price

Tasks

1. Build a quote stack for the shaped product.
2. Compare PV-only and PV+BESS dispatch value.
3. Quantify the contract-sized short/long exposure behind the shaped-product premium.
4. Identify which risks remain with the seller.
5. Decide whether to offer the shaped product, offer only as-produced energy, or no-bid.
6. State the first assumption that would change the decision.

Worked Solution

The quote stack must start with the energy the asset can naturally provide, then add the cost of reshaping that energy into the buyer's requested profile. In this scenario, the buyer asks for **100 GWh/year** flat while the solar project produces **90 GWh/year** ; the annual clean-energy shortfall and any certificate-backed replacement supply need their own line rather than being hidden inside hard-hour firming. The quote is usable only if the term sheet assigns the residual mismatch to a party, a price, a cap, or a reopener.

Component	EUR/MWh	Risk Owner If Included In Fixed Price
Raw as-produced clean energy	45.0	Seller
Replacement clean energy and certificates for annual shortfall	1.0	Seller
Shape and <u>firming cost</u>	9.0	Seller
Expected <u>imbalance</u> and forecast error	2.0	Seller / <u>BRP contract</u>
Grid, tariff, and settlement administration	1.5	Seller
Certificates and evidence handling	1.0	Seller

Component	EUR/MWh	Risk Owner If Included In Fixed Price
Credit and tenor allowance	1.5	Seller
Battery <u>opportunity cost</u>	3.0	Seller
Risk margin	2.0	Seller
Indicative shaped quote	66.0	

At 66 EUR/MWh, the product sits below the buyer's stated 68 EUR/MWh ceiling. The tail risk still needs an explicit boundary.

Dispatch And Baseload Premium Appendix

A senior quote review should test the customer promise at two levels. First, does the battery improve the asset's capture value? Second, how much of that value can be assigned to the shaped customer product after reserving dispatch rights, degradation allowance, settlement obligations, and alternative market use?

The synthetic dispatch summary gives the asset-level value pool:

Metric	PV-only	PV+ <u>BESS</u>	Commercial Use
Project output basis	90,000 MWh/year	90,000 MWh/year	Same solar asset
<u>Capture price</u>	45 EUR/MWh	54 EUR/MWh	Revenue per solar MWh
Revenue	4.05 million EUR/year	4.86 million EUR/year	Before <u>degradation</u> , fees, warranty limits, and taxes
Gross timing uplift		0.81 million EUR/year	<u>Value pool to allocate</u>

Metric	PV-only	PV+	Commercial Use
Uplift on solar output		9.0 EUR/MWh	Asset-level uplift
Uplift on buyer volume		8.1 EUR/MWh	Equivalent spread over 100 GWh/year

PV-only revenue = 90,000 MWh x 45 EUR/MWh = 4.05 million EUR/year
 PV+BESS revenue = 90,000 MWh x 54 EUR/MWh = 4.86 million EUR/year
 Gross timing uplift = 4.86 - 4.05 = 0.81 million EUR/year
 Equivalent spread over buyer volume = 810,000 / 100,000 MWh = 8.1 EUR/MWh

Treat the 8.1 EUR/MWh as a value pool that must be allocated. If the customer product consumes dispatch rights, state of charge, cycle budget, and evening discharge capacity, the value becomes an opportunity cost inside the shaped quote. If the seller retains enough control to use the battery elsewhere, the customer receives only the portion of battery value assigned to the product.

The second test prices the shape on the contract-sized promise rather than on the full project or portfolio. A full-project view can let unrelated surplus hide the specific cost of serving a flat buyer profile.

Contract-Sized Profile Test	No <u>BESS</u> Shaping	With <u>BESS</u> Shaping
Short hours	4,300 h/year	2,900 h/year
Short volume bought	18,000 MWh/year	11,500 MWh/year
Average short purchase price	76 EUR/MWh	69 EUR/MWh
Long volume sold	18,000 MWh/year	10,800 MWh/year
Average long sale price	45 EUR/MWh	54 EUR/MWh
Expected net <u>shape cost</u>	558,000 EUR/year	210,300 EUR/year

Contract-Sized Profile Test	No Shaping	With Shaping
Shape premium on buyer volume	5.6 EUR/MWh	2.1 EUR/MWh

No-BESS shape cost = $(18,000 \times 76) - (18,000 \times 45) = 558,000$ EUR/year

No-BESS shape premium = $558,000 / 100,000$ MWh = 5.6 EUR/MWh

With-BESS shape cost = $(11,500 \times 69) - (10,800 \times 54) = 210,300$ EUR/year

With-BESS shape premium = $210,300 / 100,000$ MWh = 2.1 EUR/MWh

Expected shape-cost reduction = $558,000 - 210,300 = 347,700$ EUR/year

This dispatch test strengthens the bounded-offer recommendation while keeping the collar in place. The BESS lowers expected shape cost in this scenario, while hard-hour residuals can still dominate the fixed risk margin when the system is stressed.

Numerical Intuition

The annual clean-energy shortfall is:

$$100 \text{ GWh/year} - 90 \text{ GWh/year} = 10 \text{ GWh/year}$$

If replacement clean energy and certificate handling cost 10 EUR/MWh more than the raw as-produced line in this scenario, the annual shortfall adds:

$$\begin{aligned} 10,000 \text{ MWh/year} \times 10 \text{ EUR/MWh} &= 100,000 \text{ EUR/year} \\ 100,000 \text{ EUR/year} / 100,000 \text{ MWh/year} &= 1 \text{ EUR/MWh} \end{aligned}$$

The stress firming cost is:

$$20 \text{ MWh/h} \times 250 \text{ h/year} \times 120 \text{ EUR/MWh} = 600,000 \text{ EUR/year}$$

The fixed risk margin in the quote is:

$$100,000 \text{ MWh/year} \times 2 \text{ EUR/MWh} = 200,000 \text{ EUR/year}$$

The quoted margin falls short of the illustrated stress case by 400,000 EUR/year .

$$\begin{aligned} \text{Uncovered stress cost} &= 600,000 - 200,000 = 400,000 \text{ EUR/year} \\ \text{Premium needed to absorb the full stress} &= 400,000 / 100,000 \text{ MWh} = 4 \text{ EUR/MWh} \\ \text{Full stress-inclusive fixed quote} &= 66 + 4 = 70 \text{ EUR/MWh} \end{aligned}$$

That price exceeds the buyer's 68 EUR/MWh ceiling. The shaped product is commercial at 66 EUR/MWh only if the term sheet caps, indexes, passes through, or reopens hard-hour firming beyond the quoted risk margin.

Decision Memo

Recommendation: offer the shaped product at 66 EUR/MWh only with a residual-firming collar.

The collar should be explicit. In this scenario the fixed price includes 200,000 EUR/year of risk margin. The seller can cover hard-hour firming inside that budget; volumes or costs beyond it need a pass-through, index, hard cap, or reopener. If the buyer wants the seller to carry the full illustrated stress case inside the fixed price, the quote rises to about 70 EUR/MWh, above the stated buyer ceiling.

The dispatch appendix adds evidence for the bounded offer while leaving stress risk in place. The BESS reduces the expected contract-sized shape cost from 5.6 EUR/MWh to 2.1 EUR/MWh, but the same flexibility also has gross timing value and competing uses. The quote should keep the battery opportunity-cost line unless the seller can prove that the customer's dispatch requirement and the alternative battery value do not conflict.

The investment-committee test is whether the price, cap, credit terms, evidence package, data rights, and dispatch rights describe the same product. If the buyer wants the seller to carry unlimited residual firming, the quote must move up or the product must narrow.

The as-produced fallback should remain on the table at 45 EUR/MWh. It is less valuable to the buyer, but it avoids making a fixed-price promise that the asset cannot physically support under stress.

Decision Boundary

The approval question is broader than a binary shaped-PPA vote. It is a choice among fixed shape, bounded shape, indexed residual, partner-backed firming, or as-produced fallback. The seller should approve the highest-value promise whose residual exposure is capped, evidenced, and controllable.

Asset-level dispatch value can support the product case, but the product decision should be made on the contract-sized residual exposure. Portfolio surplus, unrelated merchant upside, and average annual energy cannot erase a short-hour delivery promise.

Assumption Flip

The recommendation changes if the residual hard-hour volume rises from 20 MWh/h to 35 MWh/h.

$$35 \text{ MWh/h} \times 250 \text{ h/year} \times 120 \text{ EUR/MWh} = 1,050,000 \text{ EUR/year}$$

At that point the seller either needs a much higher fixed price, a portfolio hedge, a third-party firming partner, a softer shape, or a no-bid decision.

Failure Mode To Avoid

Approve the fixed shaped quote because the expected quote stack is below the buyer's ceiling, then leave hard-hour residuals to be absorbed by the portfolio without a cap, index, or reopener.

That decision leaves the portfolio carrying an unbounded residual exposure. The product promises shape, timing, settlement, and fallback procurement alongside annual clean energy.

A second failure is treating gross [BESS](#) capture uplift as a free customer discount. If the same MWh, cycles, state of charge, and dispatch rights are needed for the shaped profile, the uplift is part of the opportunity-cost test.

Links Back To The Lectures

- [Load Is Not Energy](#)
 - [Solar Production](#)
 - [PPA Risk Allocation](#)
 - [PPA Structure Catalog](#)
 - [PPA Pricing](#)
 - [BESS Revenue Stacks](#)
 - [Dispatch Optimization](#)
 - [Project Valuation](#)
 - [Future Of PPAs](#)
-

Case 2. The 1h, 2h, Or 4h Battery Decision

References: [7] [8] [9] [10] [11] [12] [13] [5]

Battery duration is an investment claim: the next MWh of energy capacity must earn more than it costs after efficiency, degradation, tariffs, grid rights, and contract obligations. This case asks the duration question as an investment test rather than as a technology preference.

Decision Question

For a constrained solar site, should the project build a 1-hour, 2-hour, or 4-hour battery?

Situation

A solar project has secured land and a grid position. The grid connection can export the solar plant, but adding a battery energy storage system (BESS) requires choosing a battery duration before the power purchase agreement (PPA) and optimizer contracts are finalized.

The commercial team wants a simple answer. The answer depends on how scarcity, grid access, contract value, warranty limits, and insurance limits change from the first hour to later hours.

Scenario Data

These are scenario assumptions for the case.

Item	Value
Solar project size	80 megawatts peak (MWp)
Grid export limit	60 MW
Battery power rating	30 MW
Candidate durations	1h, 2h, 4h
Carrying charge used for the test	10% of incremental capex/year
<u>Round-trip efficiency</u>	88%

Item	Value
<u>Degradation allowance</u>	included in net margin
Insurance policy	scenario assumes the modeled operating plan is insurable; if a policy caps cycles or duty cycle below the modeled plan, rerun the test under that cap

Incremental Block	Incremental Energy	Incremental Capex	Annual Hurdle
First hour	30 MWh	5.0 million EUR	0.50 million EUR/year
Second hour	30 MWh	3.5 million EUR	0.35 million EUR/year
Third and fourth hours	60 MWh	7.0 million EUR	0.70 million EUR/year

Incremental Block	Usable Cycles/Year	Net Value Per MWh Moved	Extra <u>Curtailment</u> /Contract Value
First hour	310	55 EUR/MWh	0.05 million EUR/year
Second hour	240	45 EUR/MWh	0.15 million EUR/year
Third and fourth hours	120	25 EUR/MWh	0.07 million EUR/year

Duration Gates

The duration decision needs more than an arbitrage calculation. Each incremental hour must pass the commercial, technical, and contractual gates that make the model value collectible.

Gate	What To Check	Failure Signal
Marginal value	the next MWh of energy capacity clears its own hurdle	average spread from the first hours is applied to all hours

Gate	What To Check	Failure Signal
Cycle, warranty, and insurance budget	modeled throughput, average <u>state of charge</u> , C-rate or charge/discharge rate, temperature envelope, data-retention obligation, degradation allowance, warranty terms, and insurance policy are mutually compatible	optimizer earns value by consuming life, evidence, or coverage headroom not priced in the case
Grid import and export rights	<u>connection agreement</u> permits the charging and discharging pattern	battery cannot access the hours that justify the duration
Product eligibility	extra duration qualifies for the intended reserve, firming, or <u>PPA</u> product	extra MWh exists physically but earns no contracted premium
Control rights	dispatch rights remain with the party whose economics justify the investment	<u>PPA</u> , optimizer, and reserve obligations all assume priority
<u>Augmentation</u> option	site layout, transformers, inverters, and permits leave room for later expansion	management must choose all duration now under weak evidence

A capacity warranty may care about usage/cycling degradation, calendar degradation, temperature, average state of charge, C-rate, data storage, and repair response. The case therefore treats "cycles/year" as a commercial shorthand. Before live use, replace it with the warranty-compatible operating envelope and the actual insurance-policy limits.

The product-eligibility gate should use named product requirements. In the Nordic transmission system operator (TSO) battery paper, frequency containment reserve for disturbances (FCR-D) needs at least 20 min endurance, frequency containment reserve for normal operation (FCR-N) needs 1 h in each direction, automatic frequency restoration reserve (aFRR) can require 1–4 h, and manual frequency restoration reserve (mFRR) uses a

different activation and endurance logic. Those examples are product checks for this case: 1h, 2h, and 4h should be matched to a specific revenue product before they are compared as asset labels.

Tasks

1. Calculate the annual value of each duration block.
2. Compare each block with its annual hurdle.
3. Recommend 1h, 2h, or 4h.
4. State what evidence would change the decision.

Worked Solution

The first hour value:

$$\begin{aligned} & 30 \text{ MWh} \times 310 \text{ cycles/year} \times 55 \text{ EUR/MWh} + 50,000 \text{ EUR/year} \\ & = 561,500 \text{ EUR/year} \end{aligned}$$

The first hour clears its 500,000 EUR/year hurdle by about 61,500 EUR/year.

The second hour value:

$$\begin{aligned} & 30 \text{ MWh} \times 240 \text{ cycles/year} \times 45 \text{ EUR/MWh} + 150,000 \text{ EUR/year} \\ & = 474,000 \text{ EUR/year} \end{aligned}$$

The second hour clears its 350,000 EUR/year hurdle by about 124,000 EUR/year.

The third and fourth hour value:

$$\begin{aligned} & 60 \text{ MWh} \times 120 \text{ cycles/year} \times 25 \text{ EUR/MWh} + 70,000 \text{ EUR/year} \\ & = 250,000 \text{ EUR/year} \end{aligned}$$

The third and fourth hours miss their 700,000 EUR/year hurdle by about 450,000 EUR/year.

Numerical Intuition

The best duration is the last block whose marginal value exceeds its marginal cost.

Option	Annual Value	Annual Hurdle	Margin To Hurdle
1h	0.56 million EUR/year	0.50 million EUR/year	+0.06 million EUR/year
2h	1.04 million EUR/year	0.85 million EUR/year	+0.19 million EUR/year
4h	1.29 million EUR/year	1.55 million EUR/year	-0.26 million EUR/year

The 4-hour option creates more gross value than the 2-hour option, but it destroys value at the margin in this scenario.

Decision Memo

Recommendation: build the 2-hour battery, preserve design optionality for later [augmentation](#), and do not commit to 4 hours until the site has stronger evidence of longer-duration scarcity or contracted value.

The binding constraint is the weak marginal value of hours three and four. The 2-hour asset captures the strongest short-duration price and [curtailment](#) value while avoiding a large capital block with too few high-margin cycles.

The investment decision should approve the 2-hour build and require the design team to preserve practical [augmentation](#) options where the cost is modest: space, cable routes, transformer headroom, inverter/control compatibility, and permitting language. Preserving those options keeps the project from paying today for duration that the evidence does not yet support.

The commercial team should ask for six pieces of evidence before revisiting 4 hours:

- a contract that pays for longer sustained delivery
- tariff or grid rules that reward longer import/export management
- historical or forecast price data showing enough multi-hour scarcity events after degradation and fees
- warranty, insurance, and augmentation terms showing that the value can be captured without consuming battery life or coverage headroom faster than priced
- battery management system data-access and storage arrangements strong enough to support warranty, availability, and safety disputes
- dispatch-control rights showing how PPA, reserve, merchant, and emergency uses are prioritized against the same state of charge

The rival memo should still be written. A 4-hour battery may be the better strategic move if the project can contract longer delivery, preserve scarce grid capacity, or avoid an expensive future augmentation. The third and fourth hours need their own paying scarcity; extra resilience language does not finance them.

Rival 4h Argument	Evidence Needed
the customer will pay for longer firming	signed product premium, penalty structure, and hourly shape
grid import/export rights make longer duration scarce	connection terms, tariff treatment, and <u>curtailment</u> evidence
future <u>augmentation</u> would be expensive or impossible	layout, transformer, inverter, permit, and original equipment manufacturer constraints
reserve or capacity products require longer sustain	named product rule, qualification path, and opportunity-cost test
portfolio risk justifies extra duration	portfolio stress hours and alternatives to owning the MWh

Decision Boundary

The case approves the 2-hour battery because it is the last duration block that clears its own hurdle. The committee should not reopen 4 hours because the average project economics

look better; it should reopen only when the third and fourth hours have named paying events, compatible warranty and insurance treatment, and deliverable grid rights.

Assumption Flip

The 4-hour decision changes if the third and fourth hours can earn 70 EUR/MWh net value for 160 cycles/year.

$$\begin{aligned} &60 \text{ MWh} \times 160 \text{ cycles/year} \times 70 \text{ EUR/MWh} + 70,000 \text{ EUR/year} \\ &= 742,000 \text{ EUR/year} \end{aligned}$$

It can also change in the other direction if an insurance policy caps eligible operation below the modeled cycle need. In that case, replace the modeled useful cycles with the policy-compatible cycle count before comparing duration blocks. The project may still proceed, but only after the owner obtains different coverage, changes the operating plan, or prices the lost optionality.

That clears the 700,000 EUR/year hurdle. The case for 4 hours depends on the frequency and value of longer scarcity windows, and on whether the operating plan remains compatible with warranty, insurance, tariff, and control-right constraints.

Failure Mode To Avoid

Choose four hours because the incremental capex per MWh is lower than the first two hours, then apply the average short-duration spread to all four hours.

That decision uses the economics of the valuable first hours to justify weaker later hours. The third and fourth hours need their own scarcity evidence, contract value, tariff benefit, or portfolio hedge value.

Links Back To The Lectures

- [Frequency And Reserves](#)
- [Battery Storage And Time Shifting](#)
- [BESS Revenue Stacks](#)
- [Battery Duration Selection](#)
- [Battery Degradation, Warranties, And Operating Constraints](#)

- Storage Duration And Long-Duration Flexibility
-

Case 3. Grid-Constrained Site Assessment

References: [11] [14] [15] [16] [17] [18] [19] [20] [21] [22] [23]

Solar development often begins with land and irradiation. Project value often ends with grid access, tariffs, congestion, curtailment, and what the [connection agreement](#) actually permits. The work product is a grid-aware investment assessment for three attractive sites.

Decision Question

Which site should the independent power producer ([IPP](#)) prioritize when irradiation, grid rights, tariffs, and portfolio fit point in different directions?

Situation

The development team has three candidate sites for a new solar-plus-storage project. All three can be developed, but only one can receive near-term investment resources.

The highest-yield site must still be tested against grid economics and commercial optionality.

Recent official ACER and ENTSO-E evidence strengthens the reason this is not late diligence. ACER’s 2024 infrastructure and cross-zonal monitoring reports tie congestion, [curtailment](#), cross-zonal capacity, and grid investment to market value. ENTSO-E’s TYNDP 2024 infrastructure-gaps report treats grid and storage expansion as system-cost levers. The case should therefore keep ranking sites by executable rights, not resource quality alone.

Scenario Data

These are scenario assumptions. They are not tariff quotes or connection terms. The grid and tariff examples used here support the categories to diligence: firm timing, reserved capacity, import/export rights, limited-access terms, [curtailment](#) treatment, capacity-tariff exposure, and producer-payment geography.

Metric	Site A	Site B	Site C
Solar size	100 megawatts peak (MWp)	90 MWp	110 MWp

Metric	Site A	Site B	Site C
Expected PV yield	1,060 kWh/kWp/year	1,000 kWh/kWp/year	1,030 kWh/kWp/year
Earliest firm connection	2030	2028	2028
Export limit	65 MW	80 MW	55 MW
Expected <u>curtailment</u>	5%	1%	8%
Grid/tariff/loss cost	8 EUR/MWh	4 EUR/MWh	11 EUR/MWh
Battery energy storage system (<u>BESS</u>) import allowed	conditional	yes	no
Portfolio zone fit	weak	strong	medium
Permitting complexity	medium	medium	low

Assume the merchant value before grid costs is 48 EUR/MWh for all sites. Ignore capex differences for this first-pass comparison.

Tasks

1. Calculate net annual MWh after curtailment.
2. Calculate net value per MWh after grid/tariff/loss cost.
3. Estimate annual gross margin before capex and opex.
4. Add a grid-rights and timing assessment.
5. Recommend the priority site.

Worked Solution

Annual MWh before curtailment:

Site	Calculation	Gross Output
A	100,000 kWp x 1,060 kWh/kWp	106,000 MWh/year
B	90,000 kWp x 1,000 kWh/kWp	90,000 MWh/year
C	110,000 kWp x 1,030 kWh/kWp	113,300 MWh/year

After curtailment:

Site	Net MWh/year
A	100,700
B	89,100
C	104,236

Net value per MWh:

Site	Value Before Grid Cost	Grid/Tariff/Loss Cost	Net Value
A	48 EUR/MWh	8 EUR/MWh	40 EUR/MWh
B	48 EUR/MWh	4 EUR/MWh	44 EUR/MWh
C	48 EUR/MWh	11 EUR/MWh	37 EUR/MWh

Annual gross margin before capex and opex:

Site	Net MWh/year	Net Value	Annual Gross Margin
A	100,700	40 EUR/MWh	4.03 million EUR/year
B	89,100	44 EUR/MWh	3.92 million EUR/year
C	104,236	37 EUR/MWh	3.86 million EUR/year

Grid Rights Assessment

Site	Invest / Hold / Drop	Binding Reason	First Evidence That Would Change The Treatment
A	Hold	2030 firm connection and conditional BESS import rights make the higher annual margin hard to use	firm 2028 connection, confirmed import rights, and a priced grid-cost/security package
B	Advance	2028 connection, low curtailment , full BESS import, and strong portfolio fit make the cash flow executable	unexpected capacity-tariff exposure, queue slippage, or weaker metering/import rights
C	Drop from near-term priority	no BESS import, high curtailment , and high grid/tariff/loss cost weaken shaped-product optionality	faster-than-assumed permitting path plus changed import/export rights

The tariff and connection question includes EUR/MWh, reserved capacity, geography, limited access, and producer payments. Nordic tariff work increasingly separates these items. A site assessment should therefore ask what capacity is being reserved, when it is firm, whether

limited or interruptible access is acceptable, and whether import rights let the battery charge when the commercial case requires it.

Numerical Intuition

The top-line spread between sites is small after grid effects, while the timing penalty is large.

Timing Test	Calculation	Result	Decision Meaning
Site A annual uplift over Site B	4.03 - 3.92 million EUR/year	0.11 million EUR/year	small recurring advantage
<u>Opportunity cost</u> of waiting for Site A instead of advancing Site B	2 years x 3.92 million EUR/year	7.84 million EUR	near-term margin deferred
Simple catch-up period before discounting	7.84 / 0.11 years	about 71 years	Site A does not beat Site B unless the connection and import-rights assumptions change

Use this as an undiscounted comparison rather than a project net present value (NPV). It keeps a small annual yield advantage from hiding a large timing and optionality penalty.

Decision Memo

Recommendation: prioritize Site B.

Site A has the highest simple annual gross margin, but Site B has the best decision quality. It connects earlier, has lower curtailment, lower grid cost, stronger portfolio fit, and full BESS import rights. Those features make it more useful for shaped PPAs, reserve participation, and portfolio balancing.

Site A remains a development hold, not a near-term investment priority, unless firm connection timing and battery import rights change.

The first diligence request should be the grid-rights evidence pack: firm connection date, export capacity, import capacity, connection contribution, reinforcement cost, securities or deposits, queue milestones, curtailment rules, capacity-tariff exposure, producer-payment geography, limited-access terms, metering boundary, and whether BESS operation is allowed under the same connection agreement or needs a separate approval.

The second diligence request should be the development evidence pack: site access, land title or lease route, geotechnical conditions, environmental restrictions, grid interconnection feasibility study, permitting path, construction logistics, engineering, procurement, and construction (EPC) allocation, operations and maintenance (O&M) assumptions, and resource-assessment basis. These are core bankability inputs rather than late-stage documentation.

Site A should remain in development if the connection timing improves or if a buyer values the location enough to pay for the grid uncertainty. Site C is not a priority unless its low permitting complexity creates a materially faster path than assumed.

Decision Boundary

Read the site ranking as a grid-rights decision. Site B wins because its connection, import right, and timing make more products executable. Site A becomes attractive again only when those rights change, even if its P50 generation remains higher.

Assumption Flip

The recommendation changes if Site A obtains firm 2028 connection and full BESS import rights.

Then the main penalty is grid cost rather than timing and optionality. Site A's higher output may justify prioritization, especially if the portfolio already has enough exposure in Site B's zone.

Failure Mode To Avoid

Rank the sites by base-case P50 generation value, then treat connection timing, import rights, [curtailment](#), metering boundary, and tariff exposure as diligence items.

That decision puts the grid outside the economics even though the grid determines when the MWh is deliverable, what the battery can do, and which commercial products remain feasible.

Links Back To The Lectures

- [Networks, Bidding Zones, And Tariffs](#)
 - [Voltage, Reactive Power, Inertia](#)
 - [Nordic Market Structure, Settlement, And Grid Evidence](#)
 - [Grid Connection And Tariff Economics](#)
 - [Solar Asset Cash Flow](#)
 - [Portfolio Modeling And Hedging](#)
 - [Grid Constraints And Portfolio Strategy](#)
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Case 4. Battery Revenue Stacks Under Saturation

References: [24] [25] [26] [27] [7] [9] [12] [28] [13] [5] [29]

A battery revenue stack is a set of coupled obligations on the same MW, MWh, state of charge, grid rights, cycle budget, insurance policy, and control system. When a service saturates, test which obligations remain valuable after opportunity cost and downside are counted.

Decision Question

Should the asset owner contract part of the battery’s flexibility, keep it merchant, or reserve optionality while the revenue stack evolves?

Situation

A 50 MW / 100 MWh battery can participate in energy arbitrage and reserve products. A utility offers a five-year floor contract for part of the asset. An optimizer offers full merchant dispatch with no guaranteed payment. A customer team asks whether part of the battery should be reserved for shaped PPA delivery.

The project cannot sell the same flexibility three times.

An optimizer can schedule within the conflict; it cannot remove the conflict. The dispatch model has to include market products, loss trees, interconnection limits, cycling and throughput requirements, warranty strategies, and capacity maintenance. A revenue stack is credible only if the same constraint set governs every product in the stack.

Scenario Data

These are scenario assumptions.

Item	Value
Battery	50 MW / 100 MWh
<u>Round-trip efficiency</u>	88%
Expected merchant stack value	1.70 million EUR/year

Item	Value
Downside merchant stack value	0.70 million EUR/year
Floor contract fixed payment	0.90 million EUR/year
Expected upside retained under floor	0.30 million EUR/year
Shaped PPA flexibility premium	0.75 million EUR/year
Merchant value left after reserving PPA capacity	0.80 million EUR/year
Lender-counted share of pure merchant value	20% scenario assumption
Lender-counted share of fixed floor	100% scenario assumption
Insurance cycle limit	scenario assumes no binding limit beyond the modeled cycle budget; verify policy terms before using the stack

Commercial Workbench Companion

Use the [Commercial Workbench](#) after the worked solution in Case 1. Its [BESS](#) value view mirrors this case's main tension: whether the battery should support a shaped customer product, back a floor-like contract, or keep more merchant optionality.

Tasks

1. Compare expected value and financeable value.
2. Identify which option has the strongest downside protection.
3. State the [opportunity cost](#) of reserving capacity for a shaped PPA.
4. Recommend an allocation.

Contracting Routes

Recent [BESS](#) offtake discussions use several structures rather than one generic “battery revenue” line. The case reduces them to four routes the investment committee can compare.

Route	What Is Sold	Control-Rights Consequence	When It Fits
Full merchant optimization	optimizer pursues energy, reserve, and imbalance opportunities	asset owner keeps upside and downside; lender credit is limited	owner accepts volatility and has strong controls
Floor or revenue-share contract	floor, toll, revenue swap, or minimum guaranteed payment	some upside and dispatch discretion are transferred; availability, reporting, termination, and exclusivity terms matter	project needs financeability or downside protection
Partial hybrid PPA	defined battery flexibility supports a customer profile, residual remains merchant or contracted separately	part of the battery is reserved for the PPA product	customer premium exceeds opportunity cost and risk boundary is clear
Combined hybrid PPA	renewable energy and BESS are packaged into one customer product	larger share of flexibility is locked to the customer	buyer values firmness enough to pay for it and evidence/control rights are explicit

For Nordic reserve routes, add a product-specific appendix before approving the stack. Fast frequency reserve (FFR), frequency containment reserve for disturbances (FCR-D), frequency containment reserve for normal operation (FCR-N), automatic frequency restoration reserve (aFRR), and manual frequency restoration reserve (mFRR) differ by response time, endurance, bid size, and prequalification route. The same capacity cannot be bid into conflicting obligations for the same delivery period. A revenue-stack proposal should therefore show the

exact delivery intervals where reserve capacity, activation energy, PPA delivery, arbitrage, and state-of-charge buffers overlap.

Worked Solution

Route	Expected Value	Downside Value	Financeable Value In This Scenario	Main <u>Constraint</u>
Full merchant	1.70 million EUR/year	0.70 million EUR/year	0.34 million EUR/year	Volatile and weakly financeable
Floor contract	1.20 million EUR/year	0.90 million EUR/year	0.90 million EUR/year	Gives up some upside
Shaped <u>PPA</u> plus residual merchant	1.55 million EUR/year	depends on residual firming	0.75 million EUR/year plus residual credit	Ties up control rights

The full merchant route has the highest expected value, but the lowest financeable value. The floor contract has lower expected value but stronger downside and lender readability. The shaped PPA route is attractive only if the PPA premium is high enough to compensate for lost merchant optionality.

The table drives the recommendation. The floor route gives up 0.50 million EUR/year of expected value versus full merchant, but it adds 0.56 million EUR/year of financeable value in this scenario:

$$0.90 - 0.34 = 0.56 \text{ million EUR/year}$$

The shaped PPA route has a better expected value than the floor route, but its financeable value is incomplete until residual firming, dispatch priority, and customer evidence terms are written.

Numerical Intuition

The opportunity cost of using battery capacity for the shaped PPA is:

$$\begin{aligned} & \text{full merchant expected value} - \text{residual merchant value} \\ &= 1.70 \text{ million} - 0.80 \text{ million} \\ &= 0.90 \text{ million EUR/year} \end{aligned}$$

The shaped PPA pays 0.75 million EUR/year for that flexibility. On expected value alone, it is short by 0.15 million EUR/year before considering customer value, contract tenor, and financeability.

Decision Memo

Recommendation: contract a defined tranche of the asset under a floor-like structure and keep the residual slice merchant. A first term-sheet version could reserve 25 MW / 50 MWh for the floor-backed obligation and leave the other 25 MW / 50 MWh, plus any unused state-of-charge headroom under the priority rules, available for merchant optimization. The exact split can move, but the tranche must state MW, MWh, state-of-charge band, cycle budget, priority rights, and reporting obligations. Allocate battery capacity to the shaped PPA only if the PPA premium rises or the PPA materially improves project financeability.

The binding constraint is control-right allocation. Every MW reserved for a customer profile is a MW unavailable for reserve capacity, arbitrage, or response to a new market product. The term sheet should therefore state the priority order for reserve obligations, customer shape delivery, merchant arbitrage, degradation limits, project-policy limits, minimum state of charge, availability reporting, termination triggers, and emergency override.

It should also state the evidence layer: battery management system data access, data retention, performance-test method, availability calculation, response/repair clock, and who may override dispatch during safety, grid, or warranty events. Without that evidence layer, the floor provider, optimizer, customer, insurer, and owner can each believe they own the same operating headroom.

The commercial team should price shaped PPA flexibility at its opportunity cost. If the customer cannot pay the opportunity cost, use a narrower shaped PPA plus a separate floor or optimizer agreement for the battery.

The decision pack should show two values side by side: expected value and financeable downside value. A route that maximizes expected merchant upside may still be the wrong first financing structure if it leaves the project with weak debt sizing, high collateral exposure, or no lender-readable revenue floor.

Decision Boundary

The stack should be approved by tranche, not as a single blended revenue number. Reserve the MW, MWh, state-of-charge band, cycle budget, and reporting obligation for each tranche. Anything left unallocated is optionality; anything allocated twice is an error.

Assumption Flip

The recommendation changes if merchant revenues saturate and the expected merchant stack falls by 40%.

$$1.70 \text{ million EUR/year} \times 60\% = 1.02 \text{ million EUR/year}$$

If the residual merchant value after reserving PPA capacity falls by the same 40%, it becomes:

$$0.80 \text{ million EUR/year} \times 60\% = 0.48 \text{ million EUR/year}$$

The repriced opportunity cost is then:

$$1.02 - 0.48 = 0.54 \text{ million EUR/year}$$

Against a 0.75 million EUR/year shaped-PPA premium, the PPA tranche now clears expected opportunity cost before financeability and risk adjustments. The recommendation changes only if the control-rights, cycle, state-of-charge, and insurance constraints can be written into the term sheet.

Failure Mode To Avoid

Use last year's reserve revenue, an arbitrage forecast, and a shaped-PPA premium as three additive upside lines, then haircut the total for prudence.

The same failure appears when the stack ignores project-specific policy constraints. If the actual insurance, warranty, lender, or operating documents allow fewer cycles, different duty cycles, or fewer operating modes than the optimizer assumes, part of the stack is unavailable unless the owner obtains approval, changes coverage, or changes operation.

Both versions double-count flexibility. The same MW, MWh, [state of charge](#), control rights, cycle budget, warranty envelope, data evidence, and grid connection must serve every product. Revenue stacking is constrained co-optimization; simple revenue addition double-counts the asset.

Links Back To The Lectures

- [Frequency And Reserves](#)
 - [Battery Storage And Time Shifting](#)
 - [BESS Revenue Stacks](#)
 - [Dispatch Optimization](#)
 - [Valuation Under Uncertainty](#)
 - [BESS Revenue Saturation](#)
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Case 5. Balance-Responsibility And Balancing-Service Performance Audit

References: [30] [24] [31] [32] [33] [34]

A trading or optimization partner can execute the market interface, but the asset owner still needs evidence before it changes fees, mandates, incentives, or partners. The main work is attribution. A monthly revenue gap may come from market conditions, missing data, asset constraints, forecast error, risk limits, credit limits, settlement corrections, or dispatch choices. Partner action is justified only after the evidence separates those causes.

Decision Question

Should the asset owner change partner terms now, require evidence remediation first, or treat the gap as market and constraint noise?

Situation

A renewable independent power producer (IPP) outsources balance responsible party (BRP) and balancing service provider (BSP) execution work. The partner sends monthly revenue reports. The asset owner wants to know whether the partner created value, leaked value, or operated within realistic constraints before opening a commercial renegotiation.

The partner contract allows performance review, but the current data annex is weak. Forecast files, bid files, schedule revisions, optimizer recommendations, manual overrides, meter data, activation records, and settlement corrections are delivered inconsistently.

Scenario Data

These are scenario numbers for one month. The revenue row is already net of imbalance, curtailment, and reserve effects. The lower rows are raw evidence inputs for attribution, not extra lines to add or subtract from revenue.

Metric	Actual Partner Result	Feasible Benchmark
Net energy and reserve revenue	5.90m EUR	6.40m EUR

Metric	Actual Partner Result	Feasible Benchmark
<u>Imbalance cost</u>	-0.42m EUR	-0.31m EUR
Curtailed or unavailable energy value	-0.18m EUR	-0.17m EUR
Data completeness	92%	99% required
Forecast files delivered on time	88%	98% required
Bids matching agreed risk limits	100%	100% required

The feasible benchmark uses only information available before gate closure, actual grid and asset limits, contracted control rights, agreed risk limits, and market products the partner was allowed to access. The percentage thresholds in this case are contract-scenario thresholds, not eSett rules or generic market standards.

Raw attribution pack:

Evidence Item	Observation	Initial EUR Impact
Missing or late data	forecast and bid files missing for several high-spread intervals	0.12m
Grid and asset constraints	export limit and outage periods reduced feasible dispatch	0.08m
Forecast error inside agreed range	weather-driven production miss within tolerance band	0.09m
Risk-limit conservatism	partner stayed inside owner-approved risk limits during volatile hours	0.10m

Evidence Item	Observation	Initial EUR Impact
Residual bidding/dispatch gap	unexplained difference after known constraints	0.11m

Evidence quality flags:

Required Evidence	Status	Consequence
Forecast versions and timestamps	incomplete	cannot prove what was knowable
Bid submissions and revisions	partial	cannot fully test partner action
Optimizer recommendations	missing for 18% of intervals	cannot separate tool advice from manual override
Asset availability and outages	complete	physical constraints can be attributed
Meter and settlement data	provisional, corrections pending	revenue result may move
<u>Collateral</u> and credit-limit evidence	partial	cannot tell whether otherwise feasible trades were blocked
Contract rights and risk limits	complete	feasible benchmark boundary is known

Tasks

1. Calculate the headline gap.
2. Attribute the gap using the raw evidence.
3. Decide which amount can be used for partner action now.
4. Draft a scorecard and governance recommendation.

Worked Solution

Headline gap:

benchmark_revenue – actual_revenue
= 6.40m EUR – 5.90m EUR
= 0.50m EUR

Attribution:

Driver	EUR Impact	Governability	Can Support Partner Action Now?
Missing or late data	0.12m	shared governance / contract data rights	yes, for data remediation
Grid and asset constraints	0.08m	mostly non-controllable	no
Forecast error inside agreed range	0.09m	model improvement	no, unless tolerance is renegotiated
Risk-limit conservatism	0.10m	owner policy choice	no, unless mandate changes
Residual bidding/dispatch gap	0.11m	partner execution candidate	only after evidence file is complete
Total	0.50m		

The only clean commercial action this month is data remediation. The 0.11m EUR residual may be partner execution leakage, but the missing forecast, bid, and optimizer files make a fee claim premature.

Numerical Intuition

The headline gap overstates the partner-action amount.

```
residual_bidding_gap_share = 0.11 / 0.50 = 22%
data_and_observability_share = 0.12 / 0.50 = 24%
```

Nearly a quarter of the gap is an evidence problem, and only about a fifth is a candidate execution problem. A strong scorecard prevents the owner from negotiating a 0.50m EUR accusation when the supported execution claim is closer to 0.11m EUR and still evidentially weak.

Evidence Required Before Partner Action

The owner should require raw files before using the scorecard for fee clawback, termination, incentive reset, or mandate reduction.

Evidence Layer	Required Files Or Data	What It Separates
Forecast and schedules	forecast versions, schedules, revisions, timestamps, gate-closure state	model error versus late information
Bids and dispatch	submitted bids, accepted bids, activations, optimizer recommendations, manual overrides	partner choice versus market acceptance
Asset and grid constraints	availability, outages, import/export limits, reserve commitments	controllable execution versus physical <u>constraint</u>
Meter and settlement	metered volumes, <u>imbalance</u> settlement, reserve settlement, corrections	economic result versus provisional estimate

Evidence Layer	Required Files Or Data	What It Separates
Contract rights	data rights, control rights, risk limits, permitted products	what the partner was allowed to do
Credit and <u>collateral</u>	<u>collateral</u> calls, credit limits, settlement-bank or counterparty notices	revenue result versus liquidity <u>constraint</u>
Service-level agreement (SLA) and escalation trail	dated exceptions, response times, remediation owners, and repeated defects	one-off error versus governable pattern

This mirrors the logic in Nordic settlement governance: performance and imbalance outcomes depend on reported data, settlement quantities, participant obligations, and corrections. A private partner scorecard should be at least as disciplined about raw evidence.

Commercial Workbench Companion

Use the Commercial Workbench as a compact version of the evidence discipline in this case. The partner-evidence view is useful when the question is: what evidence would make a commercial action fair and defensible?

Scorecard

Use a scorecard that separates result, evidence, and attribution.

Scorecard Area	Month Result	Status	Action
Revenue versus feasible benchmark	-0.50m EUR	watch	track with attribution; no fee action on headline gap
Data completeness	92% versus 99% requirement	fail	remediation plan with dated file-delivery obligations

Scorecard Area	Month Result	Status	Action
Forecast timeliness	88% versus 98% requirement	fail	daily timestamp feed and exception report
Risk-limit compliance	100%	pass	keep current limits until management reviews risk appetite
<u>Constraint attribution</u>	0.08m EUR	accepted	exclude from partner execution claim
Residual execution candidate	0.11m EUR	unresolved	retest after complete raw files and settlement corrections

The scorecard thresholds are scenario contract requirements for this exercise. They should be replaced by the actual service-level agreement before a live partner review. The scorecard should be repeated for at least three comparable months before commercial penalties are applied, unless the contract already defines automatic data-delivery breaches.

Decision Memo

Recommendation: do not renegotiate fees or accuse the partner based on the 0.50m EUR headline gap. Issue a formal data-rights and reporting remediation notice, agree the feasible-benchmark methodology, and put the partner on a three-month evidence watch.

Immediate actions:

- amend or enforce the data annex for forecast, schedule, bid, dispatch, optimizer, meter, activation, and settlement files
- require timestamped delivery and exception reporting
- freeze the feasible-benchmark method before seeing next month's results

- review whether owner-approved risk limits or credit limits are too conservative for volatile periods
- retest residual bidding/dispatch gaps only where evidence is complete

Partner commercial action becomes supportable if, for three comparable months:

- data completeness meets the 99% requirement
- forecast timeliness meets the 98% requirement
- settlement corrections are reconciled
- the residual bidding/dispatch gap remains material against an agreed EUR threshold or percentage of monthly benchmark revenue
- the gap occurs in intervals where action was feasible and permitted

Decision Boundary

The first enforceable decision is data remediation, not blame. The owner should escalate partner economics only after the benchmark can separate market conditions, missing data, owner limits, asset constraints, and actual partner action. That makes the scorecard harder to argue with.

Assumption Flip

The recommendation changes if complete raw files show that the partner ignored feasible profitable bids, violated the agreed strategy, or failed to act inside clear market windows while limits and asset constraints allowed action.

In that case, the 0.11m EUR residual becomes an execution claim rather than an unresolved attribution bucket, and the owner can move to fee adjustment, incentive redesign, tighter mandate, or replacement process.

Failure Mode To Avoid

Use a perfect-hindsight dispatch and attribute the full 0.50m EUR gap to the partner.

Perfect hindsight can estimate an upper bound, but it cannot govern a BRP/BSP relationship. The governance benchmark must use forecasts, market prices, liquidity, grid limits, asset status, control rights, and risk limits available at the time.

Links Back To The Lectures

- [Market Participants And Responsibilities](#)
 - [Market Clock](#)
 - [Forecast Error And Imbalance](#)
 - [Frequency And Reserves](#)
 - [Forecasting, Error, And Commercial Value](#)
 - [Trading Partners And Performance Scorecards](#)
 - [Analytical Capability Development](#)
-

Case 6. Data-Centre 24/7 Product Test

References: [1] [35] [36] [37] [4] [38] [39] [40] [41] [5] [2]

Large electricity buyers increasingly care about both when clean energy is produced and how many clean MWh are bought over a year. The seller must separate physical supply, financial settlement, certificate evidence, residual firming, credit, and control rights. The decision is whether a renewable independent power producer (IPP) should offer a 24/7-style product, narrow the promise, partner, or walk away.

Decision Question

Should the seller offer hourly matched clean power, a narrower shaped product, a partnered offer, or no-bid?

Situation

A data-centre buyer requests a clean-power product for a flat 40 MW load. The buyer asks for high hourly matching, budget stability, and clear evidence treatment.

The seller has solar assets, access to a battery project, and a trading partner. It does not own enough diverse clean generation to self-supply every hour.

IEA’s Energy and artificial intelligence (AI) report narrows the case frame. Data-centre load growth is real, but this product is exposed most directly to grid connection and stress-hour flexibility. IEA estimates that data centres account for less than 10% of global electricity-demand growth in its 2024-2030 Base Case, while around one-fifth of planned global data-centre capacity by 2030 could face grid-delay risk. Its flexibility analysis says small availability windows can matter systemically. In this case, buyer flexibility is an approval gate, not a free input.

Scenario Data

These are scenario assumptions.

Item	Value
Buyer load	40 MW flat

Item	Value
Annual demand	350.4 GWh/year
Seller-controlled clean supply	260 GWh/year
Annual clean-energy coverage	74%
<u>Hourly matching</u> before extra firming	58%
Target <u>hourly matching</u> for premium product	85%
Residual stress-hour shortfall	22 MWh/h
Stress hours	300 h/year
Firming premium in stress hours	150 EUR/MWh
Buyer premium for stronger <u>hourly matching</u>	4 EUR/MWh

Tasks

1. Calculate the annual buyer demand.
2. Calculate stress-hour firming-at-risk.
3. Compare the buyer premium with the stress exposure.
4. Define what the offer promises and how the evidence will be checked.
5. Recommend the offer shape.

Worked Solution

Annual demand:

$$40 \text{ MW} \times 8,760 \text{ h/year} = 350,400 \text{ MWh/year}$$

Stress-hour firming-at-risk:

$$22 \text{ MWh/h} \times 300 \text{ h/year} \times 150 \text{ EUR/MWh}$$

= 990,000 EUR/year

Buyer premium:

350,400 MWh/year x 4 EUR/MWh
= 1,401,600 EUR/year

The premium covers only the illustrated stress-hour procurement cost before collateral, credit, evidence handling, operational costs, margin, and any additional clean supply needed to move from the initial 58% hourly matching level toward the 85% threshold. The case still needs careful structuring before rejection or approval.

Hourly matching gap:

Matching Gap	Calculation	Result	Decision Meaning
Clean MWh currently matched hourly	350.4 GWh/year x 58%	203.2 GWh/year	starting point for the premium claim
Clean MWh needed for 85% <u>hourly matching</u>	350.4 GWh/year x 85%	297.8 GWh/year	target product obligation
Incremental time-matched clean gap	297.8 - 203.2 GWh/year	94.6 GWh/year	must come from partner supply, storage, buyer flexibility, or narrowed promise
Minimum annual clean-supply gap versus target	297.8 - 260.0 GWh/year	37.8 GWh/year	even annual volume is short before timing is solved

Premium headroom:

Premium Headroom	Calculation	Result
Buyer premium	350,400 MWh/year x 4 EUR/MWh	1.40 million EUR/year
Stress-hour <u>firming cost</u>	22 MWh/h x 300 h/year x 150 EUR/MWh	0.99 million EUR/year
Remaining headroom before credit, evidence, operations, margin, and non-stress matching	1.40 - 0.99 million EUR/year	0.41 million EUR/year
Remaining headroom spread across load	0.41 million EUR / 350,400 MWh	about 1.17 EUR/MWh

What The Offer Must Promise

The product should be split into claims that can be evidenced and controlled.

Claim Or Service	Required Evidence Or Control	Seller Exposure If Vague
Annual clean-energy coverage	generation volumes, certificates, delivery period, geography, allocation rules	buyer receives a green volume claim but not an hourly product
<u>Hourly matching</u> threshold	hourly load, hourly clean supply, storage dispatch, residual procurement records	seller may be judged against a standard it did not price
Residual firming	cap, index, pass-through, partner supply, or reopener for hard hours	uncapped procurement exposure in scarce hours
Budget stability	settlement reference, volume tolerance, <u>collateral</u> and credit terms	clean-energy product becomes an unpriced financial hedge

Claim Or Service	Required Evidence Or Control	Seller Exposure If Vague
Certificate treatment	certificate vintage, cancellation, location, matching period, audit trail	environmental claim can fail after the commercial deal is signed
Buyer flexibility	whether uninterruptible power supply (UPS), cooling, workload shifting, or on-site generation can be used and who controls it	load flexibility is assumed in the model but unavailable in operation
Reliability boundary	priority between buyer uptime, backup policy, grid services, and seller firming obligations	a flexibility resource is counted in the product when the buyer will not release it

For a data-centre buyer, flexibility resources can be valuable, but reliability constraints come first. Do not count buyer-side flexibility as seller supply unless the buyer grants defined availability, control, telemetry, settlement rights, and an uptime-compatible operating policy.

Add the accounting and certificate evidence pack before the offer is approved.

Evidence item	Source-backed reason	Contract implication
Dual Scope 2 boundary	GHG Protocol separates location-based and market-based Scope 2 reporting where contractual instruments exist	avoid language that treats annual certificates as physical hourly delivery
Energy attribute certificate status	certificates transfer attributes; attributes can be bundled or unbundled from the electricity	state whether guarantees of origin (GOs) or other instruments are included, transferred, canceled, or retained

Evidence item	Source-backed reason	Contract implication
<u>Residual mix</u> and null power	attribute-stripped energy cannot support the same renewable claim	define fallback treatment for uncovered or certificate-separated MWh
Granular time stamp	EnergyTag v2 uses UTC timestamps and production or storage-discharge intervals up to one hour	align load, clean supply, and storage evidence at the same interval
Storage evidence	storage charge and discharge records are needed to preserve attribute accounting through storage	require battery energy storage system (<u>BESS</u>) charge source, discharge record, losses, and cancellation trail

Numerical Intuition

Annual coverage can look acceptable while hourly matching is weak.

Metric	Result	Why It Matters
Annual clean-energy coverage	74%	Useful annual signal; timing exposure remains
<u>Hourly matching</u> before firming	58%	Shows residual hourly procurement need
Stress-hour <u>firming-at-risk</u>	0.99 million EUR/year	Main seller exposure in hard hours
Buyer premium	1.40 million EUR/year	Commercial headroom before other costs

Decision Memo

Recommendation: do not offer a broad self-supplied 24/7 clean-energy claim. Offer the partnered hourly-matching product only if the partner contract covers the 94.6 GWh/year time-matched clean gap, caps or passes through stress-hour procurement, and provides the evidence trail required by the selected certificate/accounting framework. Otherwise offer the narrower shaped product.

Any broad 24/7 clean-energy claim needs the applicable accounting and certificate framework in the contract. The commercial product can still be valuable if it says exactly what is being delivered: hourly matched supply to a defined threshold, financial settlement against a defined reference, and residual procurement above a defined boundary.

The recommended offer is therefore a product ladder:

Approval Gate	Minimum Condition	If Missing
Hourly matching	load, generation, storage, and residual procurement measured on the same interval	sell shaped product, not hourly claim
Residual firming	cap, pass-through, index, or named partner supply	no-bid on broad claim
Evidence framework	certificate status, cancellation rule, storage treatment, and audit trail defined before signing	remove 24/7-style language
Buyer flexibility	telemetry, control rights, availability, and uptime-compatible policy	exclude buyer flexibility from seller supply

Offer	Seller Commitment	Buyer Benefit	When To Use
As-produced plus certificates	clean energy volume and evidence	lowest complexity	buyer accepts timing exposure

Offer	Seller Commitment	Buyer Benefit	When To Use
Shaped clean product	defined hourly shape with residual boundary	budget and profile improvement	buyer values shape but not full hourly claim
Partnered hourly-matching product	threshold-based hourly matching with named firming partner and evidence pack	stronger 24/7-style claim	premium covers firming, evidence, credit, and operational cost
No-bid on broad claim	seller rejects uncapped 24/7 language	avoids mispriced obligation	offer cannot be made auditable

The decision should also record the rival memo. A seller may rationally pursue the broad product if the buyer is strategic, the relationship opens a portfolio channel, and the seller can buy the missing clean hours from a partner. That memo only survives review if it names the residual-risk owner.

Rival Claim	What Would Make It True	What Breaks It
the premium is worth the complexity	partner supply, evidence cost, credit cost, and stress-hour firming fit inside the premium	the premium only covers ordinary hours
the buyer flexibility helps the product	control rights, telemetry, availability windows, and compensation are written down	uptime policy blocks dispatch when the grid needs flexibility
the claim can be supported	certificate status, cancellation, storage evidence, and public language are agreed before signing	legal/accounting review happens after the sales promise

Rival Claim	What Would Make It True	What Breaks It
the seller should stretch for the customer	downside is capped and strategic upside is explicit	the contract turns relationship value into uncapped procurement exposure

Decision Boundary

Approve only the claim layer that the evidence can carry. Annual clean-energy coverage, shaped delivery, [hourly matching](#), flexibility service, and public claim support are different products. A strategic buyer can justify extra work, but not uncapped hard-hour procurement or undefined claim language.

Assumption Flip

The recommendation changes if stress-hour shortfall rises to 40 MWh/h.

$$\begin{aligned}
 &40 \text{ MWh/h} \times 300 \text{ h/year} \times 150 \text{ EUR/MWh} \\
 &= 1,800,000 \text{ EUR/year}
 \end{aligned}$$

That exceeds the buyer premium before other costs. The seller should then require a higher premium, a partner portfolio, a narrower matching threshold, or a no-bid decision.

Failure Mode To Avoid

Offer the product as a premium clean-power contract because annual clean-energy coverage is high, while leaving hourly matching rules, certificate evidence, and residual firming treatment to later documentation.

That decision sells the label before defining the product. [Annual matching](#) can hide the hours that drive firming cost, evidence disputes, and residual procurement exposure. A flat load needs a time-indexed delivery promise alongside the annual energy total.

Links Back To The Lectures

- [Load Is Not Energy](#)
- [Market Clock](#)

- Solar Production
 - PPA Structure Catalog
 - PPA Pricing
 - Forecasting
 - Portfolio Modeling And Hedging
 - Trading Risk, Credit, Liquidity, And Controls
 - Future Of PPAs
 - Data Centres, 24/7 Procurement, And Flexibility
-

Case 7. Year-One Energy-Management Operating Model

References: [6] [34] [42] [33] [1] [5]

An independent power producer (IPP) becomes an energy manager when it decides which revenue and risk decisions it must own, which execution activities it can outsource, what raw data it has rights to inspect, and which controls stop a model from becoming an unmanaged production process. This case forces the year-one resource allocation.

Decision Question

Given limited budget, headcount, partner data gaps, and several competing value pools, what should the company build, buy, outsource, audit, or postpone in year one?

Situation

The company has operating solar assets, a growing development pipeline, several structured power purchase agreement (PPA) discussions, and first battery projects entering commercial design. Trading execution and balancing are mostly outsourced. Partner reporting is useful but incomplete. Management wants a credible year-one operating model without approving a large desk or full energy trading and risk management (ETRM) implementation.

The operating model must support four decisions:

- shaped PPA pricing
- battery energy storage system (BESS) sizing and dispatch value
- balance responsible party (BRP), balancing service provider (BSP), and optimizer oversight
- portfolio exposure and risk escalation

Scenario Data

These are scenario assumptions.

Item	Value
Operating renewables	600 MW

Item	Value
Development pipeline under commercial review	1.8 gigawatts (GW)
First <u>BESS</u> projects	3 projects
Annual <u>PPA</u> offers under active discussion	700 GWh/year
Outsourced execution volume	500 GWh/year
Current internal analytics team	1 accountable owner + 1 analyst
Additional hire allowed	0.5 data engineer contractor
Year-one budget	900,000 EUR
External execution model	<u>BRP</u> /BSP and optimizer partners
Management <u>constraint</u>	no internal trading desk in year one

Partner data-right gaps:

<u>Data Right</u>	Current Status	Operating Consequence
Forecast versions and timestamps	partial	hard to judge what was knowable
Bid and schedule revisions	partial	partner actions cannot be fully reconstructed
Optimizer recommendations	limited	manual override versus model decision is unclear
Meter and settlement files	available with delay	monthly attribution lags
Reserve <u>activation</u> and imbalance detail	inconsistent	<u>revenue stack</u> cannot be reconciled cleanly

	Current Status	Operating Consequence
Control-right map	contract language exists, operational map missing	dispatch priority conflicts are hard to prove
Credit, <u>collateral</u> , and liquidity evidence	fragmented	cannot tell whether feasible actions were blocked by limits or cash timing

Value pools:

Value Pool	Scenario Value At Stake	Evidence Quality
Shaped <u>PPA</u> risk pricing	560,000 EUR/year	medium-high
BESS design and <u>revenue stack</u>	750,000 EUR/year	medium
Partner leakage reduction	350,000 EUR/year	low until data rights improve
Portfolio risk and hedge decisions	250,000 EUR/year	medium

Prioritize the roadmap with a simple decision-leverage score. The score is not a market fact; it forces the team to explain why a budget line deserves year-one capacity.

```

decision leverage =
  annual value at stake
  x evidence readiness
  x actionability
  / year-one spend

```

Workstream	Value At Stake	Evidence Readiness	Actionability	Year-One Spend	Score
PPA quote stack and risk map	560,000 EUR/year	0.8	0.8	160,000 EUR	2.2
BESS sizing and dispatch model	750,000 EUR/year	0.6	0.8	140,000 EUR	2.6
Partner leakage reduction	350,000 EUR/year	0.3	0.5	170,000 EUR	0.3
Portfolio risk pack	250,000 EUR/year	0.6	0.7	110,000 EUR	1.0

The data-contract work is treated as a prerequisite rather than a standalone [value pool](#). Without it, the partner-leakage score stays low because the company cannot prove which action was feasible at the time.

Control failures observed:

- commercial offers use inconsistent shape, basis, certificate, and residual-firming assumptions
- [BESS](#) cases double-count merchant and PPA use of the same MWh in some sensitivities
- partner reports show results without complete raw files
- management sees [portfolio exposure](#) after settlement rather than before decisions
- model versions and assumption owners are unclear

Tasks

1. Allocate the 900,000 EUR budget across year-one capabilities.

2. Assign headcount ownership and name what remains outsourced.
3. Prioritize a 90-day plan and a 12-month roadmap.
4. State which partner data-right gaps block partner action.
5. Define kill criteria for the roadmap.

Worked Solution

Year-one resource allocation:

Capability	Build, Buy, Outsource, Audit, Or Postpone	Budget	Owner	Reason
Data contracts and reconciliation	build/audit	130,000 EUR	accountable owner + data engineer	makes other controls usable
PPA quote stack and risk map	build	160,000 EUR	accountable owner + commercial lead	shaped offers need consistent risk pricing
BESS sizing and dispatch model	build with optimizer input	140,000 EUR	analyst + investment owner	investment committee (IC) decisions are live
Partner data-right remediation	audit/renegotiate	120,000 EUR	accountable owner + legal/commercial	raw evidence is missing
Portfolio risk pack	build	110,000 EUR	analyst + risk owner	management needs one exposure view
Forecast/vendor evaluation	buy inputs, build	90,000 EUR	analyst	value-weighted errors support

Capability	Build, Buy, Outsource, Audit, Or Postpone	Budget	Owner	Reason
layer	evaluation			multiple decisions
Feasible benchmark pilot	build small pilot	50,000 EUR	accountable owner	method first, automation later
Contingency and independent review	buy	100,000 EUR	steering group	avoids brittle models and weak validation
Total		900,000 EUR		

Postpone:

- internal trading desk
- full [ETRM](#) implementation
- automated benchmark platform across all assets
- proprietary weather model

Outsource:

- BRP/BSP [market access](#)
- 24/7 dispatch execution
- base weather/price forecast feeds
- some legal/accounting/certificate advice

Own internally:

- [PPA](#) risk allocation and quote logic
- [BESS](#) investment assumptions
- data-right requirements

- feasible benchmark methodology
- monthly portfolio risk view
- [model register](#) and assumption ownership

Forced Tradeoff

Building a polished partner scorecard immediately would spend money before the evidence layer is ready. Partner data-right gaps make a full scorecard expensive and weak. The stronger year-one allocation funds the data annex and only a narrow benchmark pilot.

Route	Spend First	What It Buys	Why It Wins Or Loses
Full scorecard now	automation and dashboards	broad reporting surface	weak attribution if raw forecasts, bids, dispatch logs, and settlement files remain incomplete
Data rights plus pilot	data annex, reconciliation, narrow benchmark	fewer assets covered at first	stronger governance because the scorecard rests on evidence the owner can inspect

The PPA [quote stack](#) and BESS model receive more year-one funding because live commercial and IC decisions can use them with available internal data. Partner commercial action waits for raw evidence.

First 90 Days

Month	Deliverable	Decision It Enables
1	data-rights inventory and gap notice to partners	shows what can be measured, audited, and renegotiated

Month	Deliverable	Decision It Enables
1	credit, <u>collateral</u> , and liquidity evidence map	shows whether trading constraints are economic, operational, or treasury-driven
1	<u>model register</u> and assumption owner map	prevents uncontrolled local models
2	shaped-PPA <u>quote stack</u> v1	quote, redesign, no-bid, or pass-through decision
2	<u>BESS</u> duration/revenue-stack model v1	1h/2h/4h, grid-right, <u>degradation</u> , and control-right decisions
3	first monthly portfolio risk pack	management view of price, shape, basis, <u>imbalance</u> , credit, liquidity, and partner exposure
3	<u>partner benchmark</u> methodology note	agrees feasible benchmark rules before automation

The first 90 days should produce usable decision artifacts, not a finished system.

Twelve-Month Roadmap

Quarter	Milestone	Acceptance Test
Q1	common <u>data contract</u> inventory	every critical feed has source, owner, timestamp, unit, granularity, validation rule, and gap owner
Q2	PPA and <u>BESS</u> models used in live decisions	at least one quote or IC recommendation changes

Quarter	Milestone	Acceptance Test
		because of the model
Q2	risk pack live	management uses it for limits, hedge questions, partner escalation, or deal approval
Q3	forecast evaluation layer	day-ahead, intraday, BESS , and partner forecast errors are value-weighted
Q3	partner data annex amended or escalation logged	missing raw evidence has a commercial owner
Q4	benchmark pilot completed	one asset class or partner has feasible attribution using time-available data

Risk Pack

The recurring risk pack should be short and decision-linked.

Section	Minimum Content	Escalation Trigger
Executive exposure	price, shape, basis, merchant tail , and liquidity summary	exposure outside approved range
PPA pipeline	quote stack , residual firming, certificates, credit, basis	unpriced risk in term sheet
BESS and dispatch	revenue stack , state-of-charge constraints, degradation, reserve/PPA conflicts	double-counted MW/MWh or cycle budget
Partner scorecard	data completeness, forecast timeliness, feasible	raw evidence below requirement

Section	Minimum Content	Escalation Trigger
	benchmark gap	
Settlement and reconciliation	meter, <u>imbalance</u> , reserve, invoice exceptions	unreconciled material variance
Credit and <u>collateral</u>	counterparty exposure, <u>collateral</u> calls, guarantees	liquidity stress or credit breach
Model changes	version changes, assumption moves, validation exceptions	unapproved model used in decision
Partner governance	missing files, late explanations, service-level agreement (SLA) breaches, unresolved incident owners	repeated defect without remediation owner

The pack is a control only if management uses it to approve, stop, redesign, hedge, escalate, or request evidence.

Data Rights Before Partner Action

Partner action is blocked where raw evidence is missing.

Action	Evidence Required First
fee reduction or clawback	complete bids, schedules, dispatch recommendations, activations, and settlement data
mandate change	proof that current limits caused avoidable loss or excess risk
optimizer replacement	evidence that feasible recommendations were materially worse than benchmark
<u>BRP/BSP</u> replacement	repeated residual execution gap after data, constraints, forecast, and risk-limit attribution

Action	Evidence Required First
incentive redesign	clear split between market conditions, constraints, and controllable partner decisions

Until those files exist, the company can demand data remediation and better reporting. It should be cautious about claiming execution leakage.

Kill Criteria

Workstream	Continue If...	Kill Or Redesign If...
PPA <u>quote stack</u>	used in live offers and changes price or structure	commercial team bypasses it or outputs cannot explain residual risk
<u>BESS</u> model	changes IC decisions or preserves valuable design option	dispatch value cannot be tied to grid rights, <u>degradation</u> , and control rights
Data remediation	partner completeness and timeliness improve against dated milestones	contract rights remain blocked after escalation
Risk pack	management takes decisions from it	it becomes a passive report
Forecast layer	<u>value-weighted error</u> improvement exceeds cost	updates arrive too late for action
Benchmark pilot	attribution uses time-available data and settlement reconciliation	it relies on hindsight or missing files

Decision Memo

Recommendation: build a small control-oriented energy-management function in year one while keeping market execution outsourced.

Approve the full 900,000 EUR allocation shown above. The first year should produce a PPA [quote stack](#), BESS design model, data-right remediation, monthly portfolio risk pack, forecast evaluation layer, and a narrow feasible-benchmark pilot. Do not approve an internal trading desk or full ETRM implementation until data rights, independent settlement checks, risk limits, and evidence-based partner attribution are working.

Add a monthly energy-risk forum with named authority for limits, credit and [collateral](#) exceptions, model approvals, partner escalations, and stop/redesign decisions. A small team can run the operating model only if decision rights are explicit.

The operating model is deliberately constrained:

- one accountable owner coordinates decisions, partner governance, and steering
- one analyst owns models, risk pack production, and value-weighted forecast analysis
- the half-time data engineer builds data contracts, reconciliation checks, and repeatable feeds
- partners continue to execute, but their raw evidence obligations become explicit

The first-year system should be deliberately small but auditable. Every model that affects a quote, dispatch view, partner scorecard, or investment recommendation needs an owner, version, source-data list, change log, and decision record. That minimum control layer should be in place before a larger trading platform or internal desk is considered.

Decision Boundary

This case funds control before ambition. The organization should expand execution only after it can observe positions, partner actions, settlement, limits, model changes, and cash exposure. Otherwise a larger operating model increases activity faster than accountability.

Assumption Flip

The recommendation changes if partners refuse or cannot provide forecast, bid, schedule, dispatch, meter, [activation](#), and settlement files on a reliable timeline.

In that case, shift more budget from model expansion to contract remediation, data interfaces, and replacement-option analysis. Building more models before the company can observe partner decisions would create false control.

Failure Mode To Avoid

Internalize market execution before the company has data rights, independent settlement checks, feasible benchmarks, risk limits, and escalation rules.

That sequence creates operational and control risk. Execution capability only helps when the organization can see the decisions being made, attribute outcomes, enforce limits, and settle disputes with evidence.

Links Back To The Lectures

- [Forecasting, Error, And Commercial Value](#)
 - [Trading Partners And Performance Scorecards](#)
 - [Analytical Capability Development](#)
 - [Data And Machine Learning In Energy Management](#)
 - [Trading Capability Design](#)
 - [IPP Operating Models For Active Portfolio Management](#)
-

Glossary

An alphabetical reference for the main terms used in the course.

ACER

European Union Agency for the Cooperation of Energy Regulators: EU agency that supports cooperation among national energy regulators and publishes EU-level monitoring, analysis, and guidance.

Actionability factor

Fraction of a model improvement that can become value after information timing, market access, liquidity, control rights, and constraints.

Actionability-adjusted uplift

Raw model or capability uplift after reducing for timing, tradable volume, liquidity, asset constraints, control rights, and organizational adoption.

Activation

Automatic or instructed conversion of reserve capability into physical response.

aFRR

Automatic Frequency Restoration Reserve: reserve automatically activated to restore frequency/control area balance after disturbance.

Aggregator

Entity that combines distributed assets or loads so they can participate commercially or technically as a portfolio.

Allocated capacity

Grid capacity assigned or reserved for a connection or network user under a specific process or agreement.

Analytical engine

Connected set of models, data flows, controls, and decision rights that turns market and asset information into repeatable commercial decisions.

Annual matching

Comparing clean-energy procurement and consumption over a year.

Apparent power

Combined active and reactive loading of AC equipment, commonly represented as S where $S^2 = P^2 + Q^2$.

As-produced PPA

PPA where buyer takes generation as produced, inheriting shape/profile exposure.

Assumption-flip test

Search for the smallest plausible change in an assumption that would reverse a recommendation.

Augmentation

Addition or replacement of battery equipment to restore or maintain usable capacity.

Back office

Function responsible for confirmations, settlements, invoicing, contract administration, reporting, and operational control records.

Backtest

Historical simulation of model or policy performance using only information available before each decision time.

Balance obligation

Obligation to plan for balance and financially settle imbalances under market rules.

Balancing capacity

MW held available for balancing service under TSO product rules, whether or not activation occurs.

Balancing energy

Energy delivered or absorbed when balancing bids or reserves are activated.

Balancing market

Real-time or near-real-time mechanism used by the TSO to maintain system balance.

Bankable revenue

Revenue treated as dependable enough for financing, depending on contract, counterparty, tenor, and downside protection.

Baseload PPA

PPA with a flat delivery profile over a defined period.

Basis risk

Mismatch between the exposure being hedged and the instrument/reference used to hedge it.

Battery duration

Battery energy capacity divided by power rating, usually expressed in hours.

Behavioral market power

Actual conduct that uses market-power potential through bidding, withholding, scheduling, or dispatch choices.

BESS

Battery Energy Storage System.

BRP

Balance Responsible Party: entity financially responsible for imbalances at a delivery point or portfolio.

BSP

Balance Service Provider: entity providing balancing services to the TSO.

C-rate

Charge or discharge rate relative to battery energy capacity; 1C roughly means full usable charge or discharge in one hour.

Calculation period

Time period over which a reference price or settlement amount is calculated.

Calendar aging

Battery degradation associated with time, temperature, and state-of-charge conditions, even without heavy cycling.

Cannibalization

Value erosion when similar generators produce at the same time and depress prices during their own production hours.

Cap

Contractual maximum price or value above which upside is limited or shared, depending on structure.

Capability priority score

Heuristic ranking of a capability by value at stake, controllability, observability, urgency, cost, and risk.

Capability-contract matrix

Build/buy/outsource table that separates internal judgment, tools, outsourced services, and non-negotiable control/data rights.

Capacity credit

Dependable capacity contribution of a resource divided by its nameplate capacity.

Capacity factor

Actual energy produced over a period divided by energy that would be produced at full nameplate output over that period.

Capacity remuneration mechanism

Market or regulatory mechanism that pays for capacity, availability, or reliability contribution in addition to energy.

Capacity warranty

Warranty or performance guarantee specifying retained usable battery capacity over time, usually conditional on operating conditions and measurement method.

Capture price

Generation-weighted average market price received by an asset or technology.

Capture rate

Capture price divided by average spot price for the same market/period.

Cash available for debt service

Cash flow available to pay scheduled debt service after operating revenue, operating costs, and allowed deductions/reserves according to the model or financing terms.

Cash Flow at Risk

Liquidity metric focused on potential cash demands versus available sources over a horizon.

Certificate contract price

Price or embedded value assigned to certificates such as GOs or EECS certificates in a PPA.

CfD

Contract for Difference: financial contract settling the difference between a strike/reference price and a market/reference price.

Closed-loop latency ratio

Feedback latency divided by decision cadence; high values mean the organization learns slower than it acts.

Collateral

Value posted to reduce credit exposure between counterparties or clearing participants.

Congestion rent

Value associated with a price difference across a constrained transmission interface multiplied by feasible flow, in simplified market-design examples.

Connection agreement

Agreement governing how an asset connects to the grid, including technical, capacity, cost, and operational conditions.

Connection charge

Charge, often one-off, covering part or all of connecting a new user or upgrading a connection.

Constraint

Physical, market, contractual, or operational limit a model or decision must obey.

Construction contribution

One-off payment or cost contribution for grid works required to connect or serve a network user.

Contract quantity

Electricity quantity specified by a PPA or related contract, such as all metered output, part of metered output, or a fixed delivery schedule.

Control right

Authority to decide, instruct, or veto scheduling, bidding, dispatch, curtailment, hedging, or reserve actions.

Cost of capital

Required return for the capital supplied to a project, including debt and equity costs.

Counterparty credit risk

Risk that the counterparty to a contract does not pay, deliver, or otherwise perform as required.

Credit support

Collateral, guarantee, parent support, letter of credit, or similar instrument used to reduce counterparty credit risk.

Cross-border PPA

PPA where buyer and generation asset are in different national markets or bidding zones, often creating basis and transmission-right questions.

Curtailment

Reduction of output below available generation, for economic, grid, or system-flexibility reasons.

Customer value

Buyer's benefit from a product, such as hedge value, budget certainty, decarbonization claim, hourly alignment, or procurement simplicity; distinct from seller cost.

Customer willingness-to-pay

Maximum price or premium a buyer may rationally accept based on hedge value, budget

certainty, decarbonization value, evidence needs, and procurement alternatives.

Cycle aging

Battery degradation associated with charge-discharge use, including throughput, depth of discharge, rate, and operating pattern.

Daily risk pack

Recurring report connecting positions, MtM, market risk, credit exposure, liquidity/collateral, basis/shape, partner execution, limits, data quality, and actions.

Data contract

Definition of a data feed's source, owner, timestamp, unit, granularity, validation rule, and decision use.

Data right

Right to receive data needed to reproduce, audit, and govern decisions.

Day-ahead market

Auction before delivery day where hourly or sub-hourly electricity volumes are cleared for the next day.

Debt

Money lent to a project or company, usually paid before equity and therefore usually cheaper but less tolerant of downside cash-flow weakness.

Debt sculpting

Shaping debt service over time to fit the cash-flow profile a lender is willing to count.

Debt sizing

Process of estimating how much debt a project can support from lender-counted cash flow, coverage requirements, tenor, interest rate, reserves, and amortization structure.

Decision latency

Time between the arrival of useful information and the latest moment a feasible action can still be taken.

Decision memo

Short recommendation document stating the decision, core evidence, retained risk, binding assumptions, owner, next measurement, and assumption-flip test.

Decision owner

Person or function accountable for accepting, rejecting, or overriding a model recommendation or operational action.

Decision variable

Quantity the optimization model is allowed to choose.

Degradation

Loss of battery capacity or performance over time due to calendar aging and cycling.

Degradation-adjusted spread

Discharge price minus charging cost adjusted for efficiency, degradation, fees, tariffs, and opportunity cost.

Delivery schedule quantity

Fixed quantity or quantities to be delivered or settled during defined periods under a contract schedule.

Depth of discharge

Fraction of stored energy practically used in a discharge.

Dispatch

Operational choice of when/how much an asset charges, discharges, generates, or bids.

Dispatchable

Controllable by an operator within technical and fuel/resource constraints.

Distribution

Lower-voltage network delivery between the transmission system and end users / distributed resources.

Diversification

Risk reduction from imperfectly correlated exposures rather than asset count alone.

DSCR

Debt service coverage ratio: cash available for debt service divided by required debt service over a period.

DSO

Distribution System Operator: entity responsible for operating a distribution network.

Duration hedge

Extra battery energy capacity bought partly to handle uncertainty in spreads, delivery obligations, reserve calls, curtailment windows, or grid constraints.

Economic withholding

Offering capacity at a price high enough that it is unlikely to clear, even though it is technically available.

Energy

Quantity accumulated or consumed over time, usually Wh, kWh, MWh, TWh.

Energy attribute certificate

Instrument that represents attributes of generated electricity, such as renewable attributes, and can be transferred or canceled separately from physical electricity depending on the scheme.

Energy management

Capability connecting forecasts, market positions, dispatch, contract obligations, imbalance exposure, partner execution, and portfolio risk.

Energy-limited

Able to provide power only until a stored-energy, fuel, water, or demand-flexibility constraint binds.

Energy-management operating system

Set of recurring decisions, data flows, models, limits, partner controls, feedback loops, and review rhythms that turns assets and contracts into governed commercial action.

Energy-only market

Market design where investment recovery relies mainly on energy and ancillary-service revenues, including scarcity prices.

Equity

Ownership capital that receives residual cash flows after operating costs, debt service, reserves, and other senior claims.

Equivalent full cycle

One full charge-discharge equivalent measured by throughput relative to usable energy capacity; partial cycles can add up to equivalent full cycles depending on the convention used.

ETRM

Energy trading and risk management system used to record, value, monitor, and control trades, positions, limits, settlement, and reporting.

Expected shortfall

Average loss conditional on losses exceeding a selected tail threshold such as VaR.

Expected unserved energy

Expected MWh of demand not served because supply is insufficient.

Exposure cube

Portfolio map that classifies risk by geography, technology, contract, hedge, flexibility, counterparty, liquidity, and data/control rights.

Fault level

Measure related to system strength and available fault current at a network point.

FCR

Frequency Containment Reserve: reserve intended to contain frequency deviations quickly after an imbalance.

Feasible counterfactual

Simulated action path that could have been taken with contemporaneous information, constraints, market access, risk limits, and control rights.

Feedback latency

Time between a decision and the reliable measurement or settlement data needed to evaluate it.

Final position

Market position carried into delivery after accepted trades, nominations, or adjustments.

Financeability lens

View of project cash flows focused on resilient cash for debt and covenant/downside requirements alongside equity upside.

Firming cost

Cost of transforming variable renewable output into a firmer or more useful delivery shape through storage, portfolio netting, market purchases, hedging, or flexibility.

Firming-at-risk

Stress measure for the cost of residual firming exposure: residual MWh in hard hours multiplied by the relevant firming premium or stress price.

Flexibility potential

Technical or practical ability of a load, generator, storage asset, or portfolio to change injection/withdrawal in response to grid or market needs.

Flexible connection agreement

Connection or network-use arrangement that allows limited, conditional, or interruptible injection or withdrawal, often to accelerate connection or reduce reinforcement needs.

Floor

Contractual minimum price or value below which settlement is protected, depending on structure.

Forecast-based dispatch

Dispatch optimized using information available at decision time.

FPA

Flexibility Purchase Agreement or flexibility-style arrangement; exact meaning must be defined by source/context before use.

Front office

Commercial function that originates, structures, prices, trades, dispatches, or manages market-facing portfolio decisions.

Generation adequacy

Generation-resource component of system adequacy.

Governance calendar

Recurring cadence for position review, partner performance, PPA pricing, portfolio exposure, model assumptions, capability strategy, and investment gates.

Granular certificate

Certificate or attribute-tracking concept that adds temporal and sometimes locational granularity to energy attribute tracking.

Grid tariff

Charge paid for use of the grid, potentially including fixed, capacity, energy, loss, and reactive-power components.

Grid-following inverter

Inverter that synchronizes to an existing grid voltage/frequency reference.

Grid-forming inverter

Inverter controlled to establish or support voltage/frequency reference behavior more actively.

Guarantee of Origin

European energy attribute certificate used to document electricity attributes within the relevant issuing and cancellation rules.

Hedge-quality checklist

Test of whether a hedge matches the underlying exposure by reference price, volume, shape, tenor, basis, liquidity, collateral, and governance.

Hourly matching

Comparing clean supply and consumption within each hour.

Imbalance

Difference between scheduled/nominated and actual generation or consumption.

Incremental duration

Additional MWh added to a battery design while keeping MW approximately fixed.

Incremental duration value

Additional value from adding MWh to a battery after additional revenue, capex, degradation, tariffs, and opportunity cost.

Independent price validation

Process of checking marks, curves, or valuations independently from the team that originated or owns the position.

Inertia

Resistance of rotating mass to sudden frequency change.

Insurance operating covenant

Condition in an insurance policy or endorsement that limits how an asset may be operated while keeping coverage, pricing, or claim eligibility intact. For BESS this may include cycle, throughput, duty-cycle, safety, monitoring, or approved-use restrictions.

Intraday market

Market after day-ahead clearing where participants adjust positions closer to delivery.

Investment-case waterfall

Path from gross physical value to equity cash flow that shows capture, curtailment, contract, grid, BESS, opex, debt, and residual effects layer by layer.

IPP

Independent Power Producer: company that owns or operates generation assets and sells electricity or related products without being the incumbent utility.

IRR

Internal rate of return: discount rate that makes project NPV equal zero.

Limited grid access

Connection or access arrangement where injection or withdrawal is constrained under specified conditions.

Liquidity at Risk

Risk measure for liquidity pressure under market, credit, collateral, or trading-liquidity stress.

Liquidity risk

Risk that a firm cannot trade, exit, or fund positions when needed without unacceptable cost or disruption.

Load

Electrical demand at a point, portfolio, zone, or system over time.

Load shape

Pattern of demand across time.

Location-based Scope 2 method

Scope 2 accounting method based on average emissions factors for the grid or location where electricity consumption occurs.

Long-duration energy storage

Storage intended to address longer scarcity periods than short intra-day shifting.

Loss charge

Charge or adjustment intended to recover electrical losses associated with network use.

Loss of load expectation

Expected duration of insufficient supply over a planning horizon.

Loss of load probability

Probability that available resources cannot meet load in a specified period.

Loss tree

Engineering model that tracks energy losses through a renewable, storage, or hybrid system so dispatch and valuation use deliverable energy rather than ideal nameplate energy.

Margin call

Demand to post additional collateral after exposure or collateral requirements change.

Marginal duration hurdle

Annualized cost and opportunity-cost threshold that the next MWh of storage energy capacity must clear before extra duration is worth buying or building.

Marginal risk contribution

Incremental stress, variance, liquidity, or concentration added by the next asset, hedge, contract, or partner dependency.

Market access

Ability to participate in a market through required contracts, systems, qualifications, collateral, and operations.

Market monitoring

Monitoring of bids, outages, constraints, behavior, and outcomes to distinguish normal scarcity, opportunity cost, and problematic conduct.

Market power

Ability to profitably influence market price away from the competitive level for a meaningful period.

Market risk

Exposure to losses from movements in prices, spreads, volatility, basis, or correlations.

Market time unit

Time granularity of traded or settled energy, such as hourly or 15-minute periods depending on market and date.

Market-based Scope 2 method

Scope 2 accounting method based on contractual instruments such as energy attribute certificates, qualifying PPAs, supplier-specific emission rates, and residual mix where available.

Merchant tail

Post-contracted or partially uncontracted period where project revenue depends more directly on market prices and exposed risks.

Merchant upside

Uncontracted or weakly contracted expected value from market optimization.

mFRR

Manual Frequency Restoration Reserve: reserve manually or centrally activated to restore

balance and manage system needs.

Middle office

Independent or semi-independent function that measures risk, validates prices, monitors limits, benchmarks performance, and challenges front-office decisions.

Missing money

Concern that market revenues may be insufficient to finance needed capacity because scarcity or investment signals are muted, risky, capped, or incomplete.

Mixed-integer linear programming

Optimization method where some variables are continuous and others are integer or binary, useful when dispatch choices include on/off, mutually exclusive, or commitment-style decisions.

Model inventory

Register of models, owners, purposes, inputs, outputs, limits, validation status, and retirement criteria.

Model register

Inventory of models, owners, inputs, outputs, validation status, and decisions supported.

Model value at risk

Scenario estimate of value exposed to a wrong model-driven action across affected MW, duration, adverse price error, and bad-event count.

Model-risk charge

Scenario or governance allowance for losses, overrides, monitoring, and control burden introduced by relying on a model.

Multi-buyer PPA

PPA where several buyers aggregate demand to contract with one generator or supplier structure.

Nameplate capacity

Rated maximum power under defined conditions.

NEMO

Nominated Electricity Market Operator: exchange/operator designated for day-ahead and intraday market coupling functions.

Netting

Partial cancellation of positive and negative deviations inside compatible commercial or settlement boundaries.

NPV

Net present value: discounted value of future cash flows minus investment cost.

Null power

Electricity from which energy attributes have been separated, so it cannot support the same market-based renewable claim without a valid contractual instrument.

Numerical weather prediction

Physics-based weather model used as an input to power, load, price, and system forecasts.

Objective function

Mathematical expression a model seeks to maximize or minimize.

Offtaker

Buyer of electricity or environmental attributes under a commercial contract such as a PPA.

Opportunity cost

Value of the best feasible alternative action forgone when an asset is committed to a different action.

Paid-or-required capability gap

Difference between system-service capability that is required or valuable at a site and capability that is already qualified and available.

Partner benchmark

Feasible counterfactual used to compare actual BRP/BSP/optimizer performance with actions available under real constraints and rights.

Pay-as-produced PPA

PPA where the buyer purchases electricity as generated, so the buyer generally takes more generation-profile exposure.

Peak load

Maximum load over a defined period and geography.

Perfect foresight dispatch

Dispatch optimized with actual future information; useful as an upper bound for operational benchmarks built from tradable information available at decision time.

Physical withholding

Making capacity unavailable even though it could physically produce.

Pivotal supplier

Supplier whose capacity is needed to meet demand under the relevant constraints.

Portfolio exposure

Sensitivity of portfolio value or cash flow to a market, physical, contractual, credit, or operational driver.

Portfolio PPA

PPA backed by a portfolio of assets rather than a single identified plant, potentially spreading resource and operational risks.

Potential future exposure

Forward-looking estimate of how large counterparty exposure may become under future market movements.

Power

Rate of energy flow, usually measured in W, kW, MW, or GW.

PPA

Power Purchase Agreement: contract for purchase/sale of electricity, often long-term.

Private-wire PPA

PPA where generation and consumption are connected through a direct/private wire rather than only through ordinary grid supply arrangements.

Product-readiness score

Heuristic assessment of whether a firm has the assets, portfolio, data, control rights, risk limits, and evidence needed to offer a more advanced PPA product.

Project policy constraint

Project-specific insurance, lender, warranty, or operating-policy limit on cycling, throughput, duty cycle, approved revenue modes, data retention, or claim eligibility.

Qualification chain

Sequence that turns technical capability into commercial value: specify, test, qualify, control, measure, settle, and enforce.

Quote stack

Pricing structure that decomposes an offered PPA price into raw energy, shape/firming, imbalance, grid/tariff, certificates, credit, tenor, optionality, and risk-margin components.

Reactive power

Component of AC power associated with voltage support and electric/magnetic fields, commonly measured in var/MVAr.

Reference price

Market price, index, or formula used to settle a financial PPA, CfD, floor/cap, or indexed price component.

Regional upscaling

Converting asset-level forecasts or samples into a portfolio, bidding-zone, or control-area forecast.

Reserve capacity

MW held available for possible TSO activation under product rules.

Reserve headroom

MW or MWh capability held back so a battery can deliver a reserve obligation if called.

Residual demand

Demand left for a supplier after subtracting other available supply and feasible imports.

Residual load

Load remaining after subtracting variable renewable generation or another specified supply category.

Residual mix

Emissions or attribute mix left for electricity consumption after contractual instruments such as certificates have been removed or claimed.

Residual obligation

Hourly MWh, EUR, evidence, or operational exposure left after contracted clean supply, storage, hedges, certificates, and control actions have been applied.

Residual Supply Index

Market-power diagnostic comparing supply available without a supplier to demand; values below one indicate pivotality in the simple form.

Retained-risk table

Decision-memo table naming risks that remain after the recommendation, their owners, warning signals, and mitigations.

Revenue quality ladder

Course tool ranking modeled value by how confidently it can support fixed obligations versus equity upside or unproven optionality.

Revenue stack

Set of revenue sources a flexible asset may pursue, subject to mutual exclusivity, opportunity cost, and operational constraints.

Risk limit

Quantitative or qualitative boundary on positions, losses, exposures, stress losses, collateral needs, or permitted strategies.

Round-trip efficiency

Energy discharged divided by energy charged over a cycle.

Scarcity duration

Time length of the shortage or obligation a resource must cover.

Scarcity pricing

High pricing that reflects real scarcity of energy, capacity, flexibility, or network capability.

Scarcity-duration distribution

Distribution showing how often scarcity or obligation events of different lengths occur, for example 1-hour, 4-hour, overnight, multi-day, or seasonal events.

Security of supply

Broad ability of the power system to serve demand reliably across operating and planning horizons.

Shadow price

Marginal value of relaxing a binding constraint by one unit in an optimization problem.

Shadow trading

Counterfactual simulation of feasible trading/dispatch decisions used to benchmark actual execution or partner performance.

Shape cost

Cost of converting asset output into a promised contract profile, including market purchases, storage losses, degradation, hedging, balancing, grid/tariff costs, and risk margin.

Shape risk

Risk that production or consumption timing differs from the contracted or hedged delivery shape.

Short-circuit ratio

Grid-strength indicator comparing available short-circuit power with connected converter capacity; detailed definition depends on method.

State of charge

Battery energy currently stored, usually expressed as MWh or percent of usable capacity.

Storage discharge granular certificate

Granular certificate associated with storage discharge, requiring matching storage charge/discharge records and loss treatment under the relevant scheme rules.

Stress test

Scenario analysis of severe but plausible or intentionally challenging market, credit,

liquidity, or operational conditions.

Structural market power

Market-power potential arising from market structure, location, ownership, concentration, constraints, or pivotality.

Subscribed capacity

Capacity level contracted for grid use or tariff purposes.

Svenska kraftnät

Swedish transmission system operator responsible for the national transmission grid and power-system operation in Sweden.

System adequacy

Longer-term ability of the system to have enough resources and network capability to meet demand under credible conditions.

System margin

Difference between available dependable resources and demand or required reserve/adequacy target, depending on definition.

System security

Short-term ability of the power system to remain balanced and withstand disturbances.

System-service option

Capability that may create value through connection approval, compliance, avoided curtailment, paid services, or future grid requirements, but is not yet base-case revenue.

Terminal value

Value assigned to the final state of a storage model, often to avoid artificial emptying at the end of the optimization horizon.

Termination amount

Amount due on early contract termination, often linked to mark-to-market or alternative valuation methods.

Throughput budget

Allowed or planned cumulative charged/discharged energy over a period or asset life.

Transmission

High-voltage bulk power transport over the main grid.

TSO

Transmission System Operator: entity responsible for secure operation of the transmission system and system balance within its mandate.

Use-of-network charge

Recurring charge for grid use, potentially covering infrastructure, losses, system services, metering, or reactive charges depending on tariff design.

Value at Risk

Loss threshold for a specified probability and horizon; it does not describe the severity of losses beyond the threshold.

Value of lost load

Estimate of consumer willingness to pay to avoid involuntary interruption of electricity supply.

Value pool

Source of possible economic improvement, such as better pricing, reduced imbalance, improved dispatch, or reduced leakage.

Value-duration curve

Curve or table showing the marginal value of adding each extra hour of usable storage duration after efficiency, degradation, grid, tariff, contract, and opportunity-cost constraints.

Value-weighted error

Forecast or model error weighted by the economic consequence of being wrong in that interval or scenario.

Virtual battery

Controllable demand or process flexibility that can absorb, defer, or release electricity demand over time in a way that mimics some storage services without storing electricity electrochemically.

Voltage

Electrical potential difference; grid voltage must remain within operational limits.

WACC

Weighted average cost of capital: combined cost of debt and equity after weighting by capital structure and tax treatment where applicable.

Warranty envelope

Operating conditions under which a battery's warranty, performance guarantee, availability commitment, or remedy structure remains valid.

Weather-dependent

Output materially depends on weather conditions outside operator control.

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